🌐 Network Scanner & CVE Finder — User Guide

🌟 Overview

This tool performs two powerful tasks:

1. 🔍 Scans open ports on a target system
2. 🛡️ Fetches known vulnerabilities (CVEs) related to services

It combines network scanning with vulnerability intelligence using data from the National Vulnerability Database

🚀 Features

- 🌐 Port scanning (1–1024 range)
- 🔓 Detects open ports and services
- 🧠 CVE lookup for known vulnerabilities
- ⚡ Fast scanning using Nmap
- 📡 API-based vulnerability intelligence

📥 Getting Started

1. Install Dependencies

pip install python-nmap requests

2. Install Nmap (Required)

This tool depends on
Nmap

👉 Download from: https://nmap.org/download.html
👉 Make sure it is added to your system PATH

3. Run the Tool

python your_file_name.py

🧭 How to Use

🔍 Step 1: Scan Target

You will be asked:

Enter target IP address:

Example:

192.168.1.1

👉 The tool will:

- Scan ports **1–1024**
- Display open ports and their state

🛡️ Step 2: Search Vulnerabilities

Next, enter a service name:

apache
nginx
openssh

👉 The tool will:

- Query CVEs from the NVD database
- Show top vulnerabilities

🔍 Output Explained

Port Scan Output

Port: 80 (tcp) - State: open

👉 Means:

- Port 80 is open
- Likely running a web server

CVE Output

CVE-2023-XXXXX: Description...

👉 Each entry includes:

- CVE ID
- Vulnerability description

🔐 Important Notes

- Requires **internet connection** for CVE lookup
- Uses **NVD API key** (keep it private)
- Scanning external systems may require permission

⚠️ Legal & Ethical Warning

This tool must only be used:

✅ On systems you own
✅ On systems you have permission to test

❌ Do NOT use on:

- Random public IPs
- Unauthorized networks

Unauthorized scanning can be illegal.

🛠️ Troubleshooting

❌ Nmap not found

- Ensure Nmap is installed
- Add it to system PATH

❌ No scan results

- Target may be offline
- Firewall may block scan

❌ API errors

- Check internet connection
- Verify API key

🎯 Use Cases

- Network security assessment
- Learning ethical hacking
- Identifying exposed services
- Checking vulnerabilities in software

🧠 Security Insight

Open ports = possible entry points

If a service has known CVEs:
👉 It becomes a **real attack surface**

📝 Final Note

This tool doesn't break into systems...

It simply listens —
to open doors, exposed services, and known weaknesses whispered in public databases.

Use it wisely, and it becomes a lens into the hidden structure of networks.